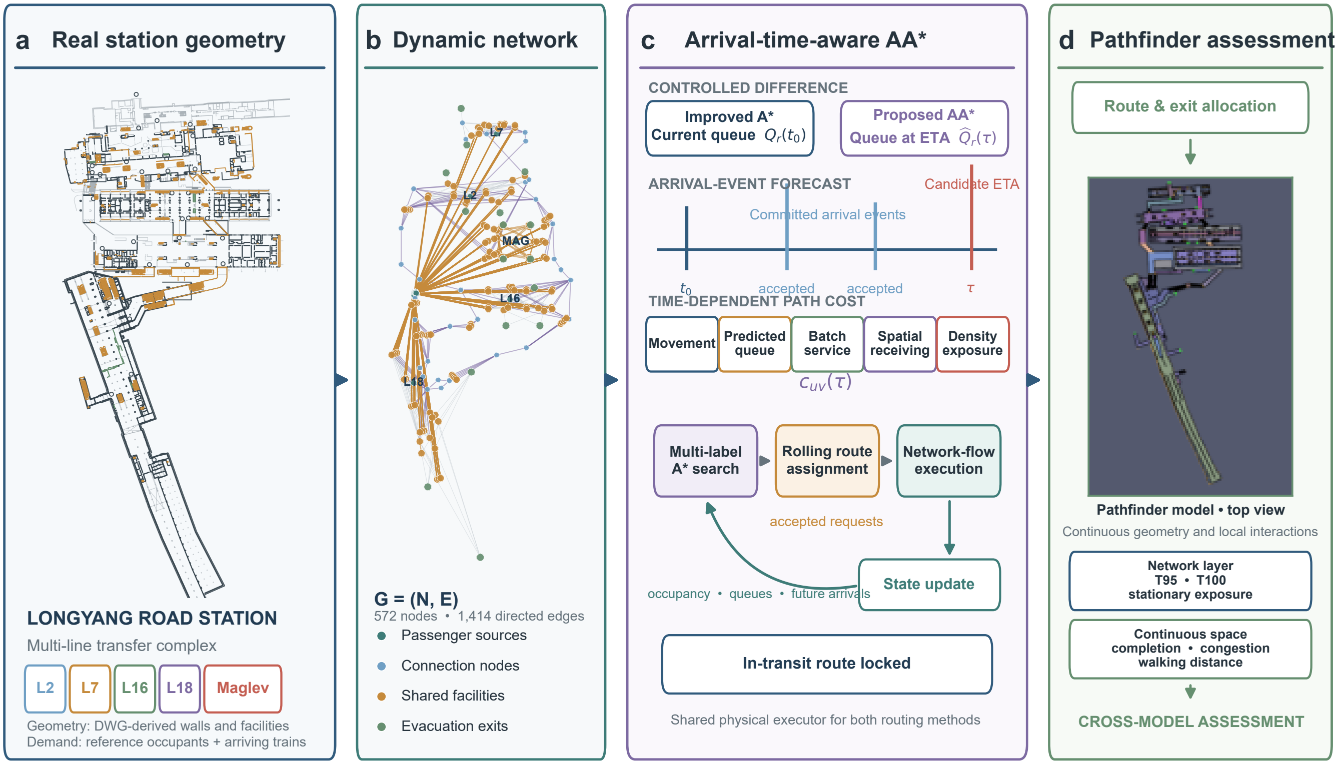
a Real station geometry
LONGYANG ROAD STATION
Multi-line transfer complex
L2
L7
L16
L18
Maglev
Geometry: DWG-derived walls and facilities
Demand: reference occupants + arriving trains
b Dynamic network
L7
L2
MAG
L16
L18
G = (N, E)
572 nodes • 1,414 directed edges
Passenger sources
Connection nodes
Shared facilities
Evacuation exits
c Arrival-time-aware AA*
CONTROLLED DIFFERENCE
Improved A*
Current queue $Q_r(t_0)$
Proposed AA*
Queue at ETA $\widehat{Q}_r(\tau)$
ARRIVAL-EVENT FORECAST
Candidate ETA
Committed arrival events
t_0
accepted
accepted
τ
TIME-DEPENDENT PATH COST
Movement
Predicted queue
Batch service
Spatial receiving
Density exposure
$c_{uv}(\tau)$
Multi-label A* search
Rolling route assignment
Network-flow execution
accepted requests
State update
occupancy • queues • future arrivals
In-transit route locked
Shared physical executor for both routing methods
d Pathfinder assessment
Route & exit allocation
Pathfinder model • top view
Continuous geometry and local interactions
Network layer
T95 • T100
stationary exposure
Continuous space
completion • congestion
walking distance
CROSS-MODEL ASSESSMENT